Suction Device for Removal of Spotted Lanternflies

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PROBLEM & CONTEXT

Problem
- Spotted Lanternflies (SLF) damage grape plants in vineyards and contaminate grape harvests
- Adult SLF removal methods damage vines, contaminate grapes, or are inefficient
  - Chemical methods, such as pesticides, can harm grape plants and impact grape growth
  - Methods of removal that involve physical contact with the grape plants can harm the plants
  - Manual methods of removal are inefficient

Goal
- Effectively capture SLF during grape growth periods without significantly damaging grape plants

Needs/Constraints
- No-contact removal method
- Easily integratable into existing agricultural practices

Impact
- Increased grape yield
- Reduced chemical use
- Lower labor costs

SOLUTION DIRECTION

Preliminary Prototype

Design Components
- Square cone
- Intermediate chamber with a moving wall (actuated by a string) to kill and contain SLF
- Collection box under chamber for SLF collection

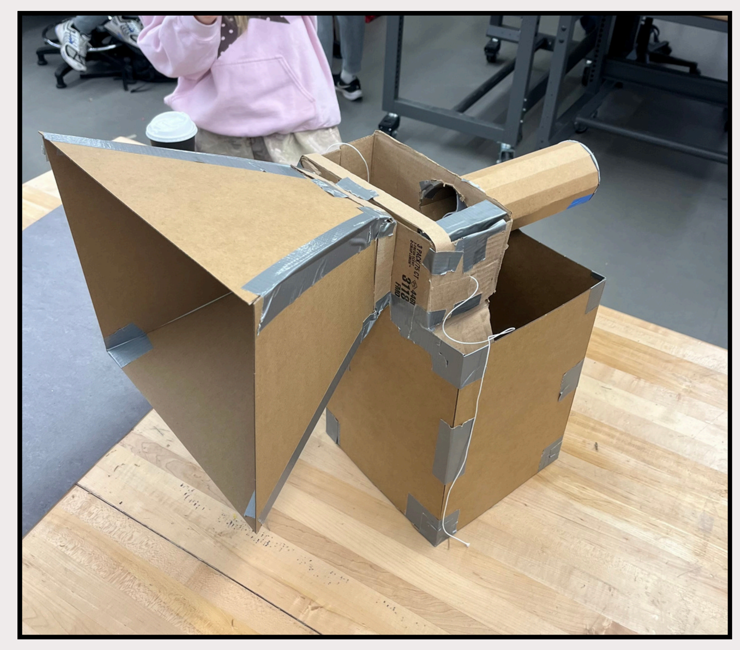

Fig. 1: Preliminary Prototype

Refined Prototype

Design Components
- Rectangular cone to maximize coverage of grape vines
- Intermediate chamber with two holes to collection chamber

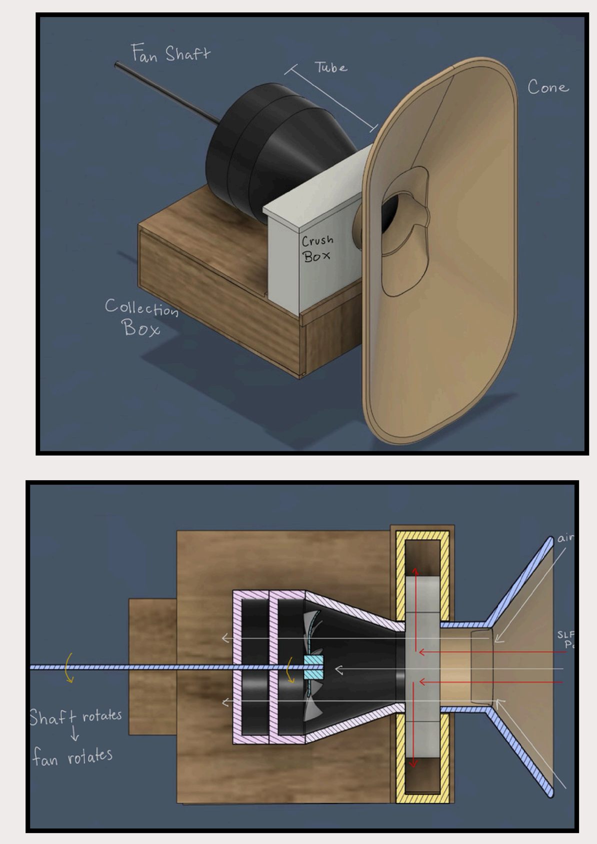

Fig. 2: Refined Prototype

Final Prototype

Design Components
- Motor and impeller to create vacuum
- Circular cone to evenly direct air and SLF into device
- Mesh cone to clear SLF from path of airflow
- Power source

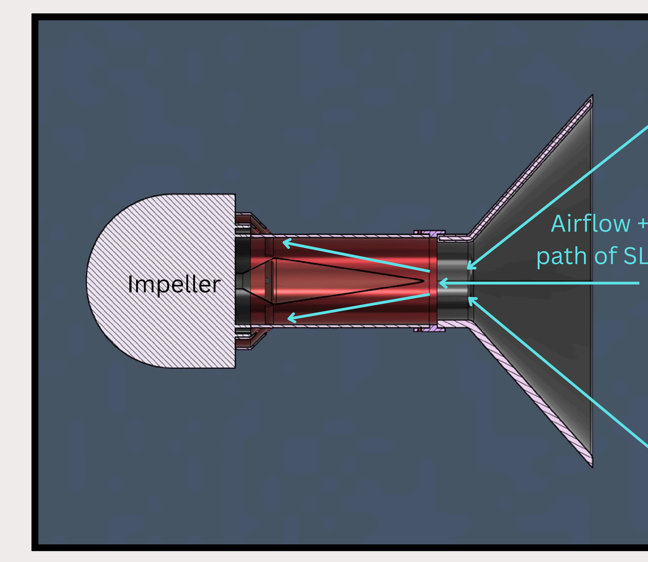

Fig. 3: Final Design Functionality

TESTING AND DESIGN VALIDATION

Testing

Longevity
Conducted testing to determine device endurance over prolonged operation by continuously operating the device for 30 minutes to test the charge of the battery and if there was any deterioration in performance over time.
Results: The device operated succesfully for the duration of the test, with no notable impacts to performance. The battery lasted the 30 minutes and the device had no variation in its performance over the period.
Conclusion: We expect no issues to arise regarding the longevity of the device. It has few moving parts to sieze up or malfunction during operation.

Debris Tolerance
Tested intake of various materials ranging from 1-5cm in diameter and 1-10g in mass.
Results: No debris caused any visible damage to the internals of the device, though many smaller and more massive debris (>5g) got trapped within the device before reaching the storage container.
Conclusion: The device will not be damaged by any SLF or plant matter that it captures.

Mounting Load
Tested strength of mounting geometry by adding mass to the device while it supported itself.
Results: Successfully supported 500g of added mass without critical failure (fracture).
Conclusion: The device is robust enough to be mounted with a full storage of SLF.

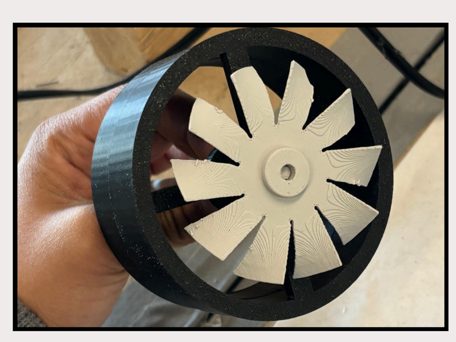

Fig. 4: Testing of a propeller to generate suction, which performed poorly.

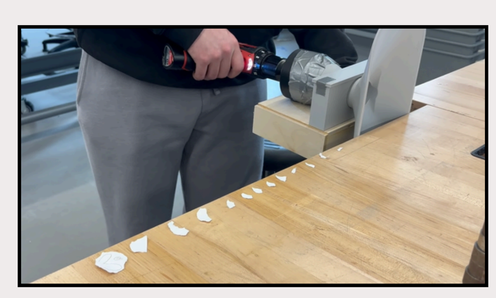

Fig. 5: Testing of the SLF size that the device can collect

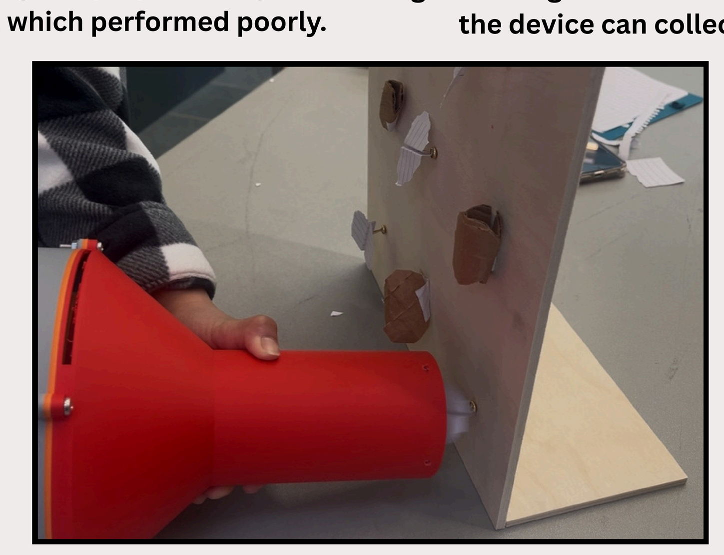

Fig. 6: Testing of the final prototype with the impeller and without the cone.

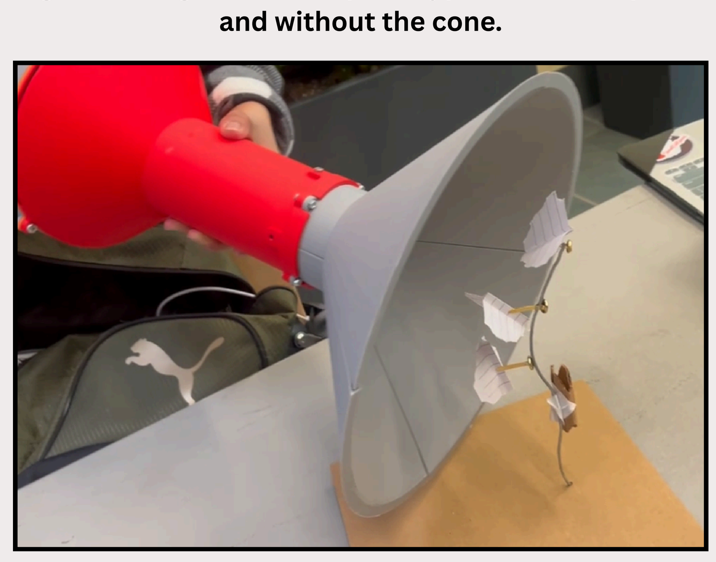

Fig. 7: Testing of the final prototype with the impeller and with the cone

FINAL PROTOTYPE

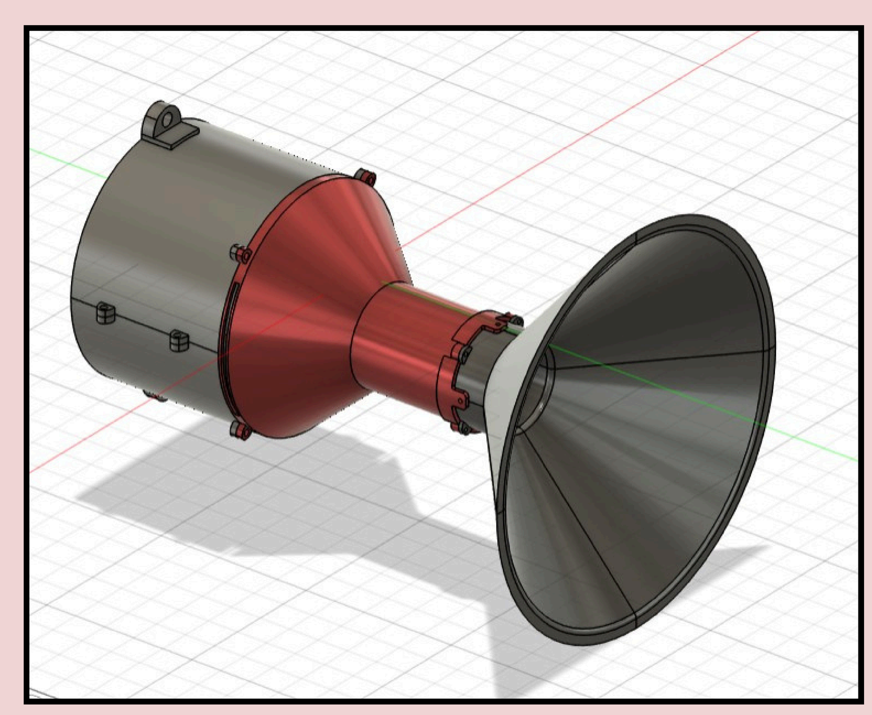

Fig. 8: CAD Model of Final Prototype

Fig. 9: Final Prototype

Key Design Improvements
- Optimized cone geometry to improve airflow and SLF capture efficiency
- Replaced propeller with impeller to reduce damage from debris and improve durability
- Integrated sloped mesh filter to block large debris while allowing SLF passage
- Designed motor casing with mounting panel so device can be easily attached to a tractor, making it easier for agricultural workers to incorporate into their existing work practices.
- Changed power source to rechargeable battery that can be recharged in between tractor expeditions

DISCUSSION

Performance

The device accomplished some of our design goals and did not fulfill others. It does remove SLF without coming into physical contact with the plants and can be easily attached to a tractor. However, the suction device had to be held quite close to our test vine, and the test SLF were only pulled off of the vine when they were positioned directly in the center of the nozzle.

The device performed better when the cone was not attached than when it was, likely due to improved airflow and a decrease in the minimum distance from the vacuum to the SLF. When the cone was attached, it took much longer to remove one SLF from the vine. When the cone was not attached, all three SLF could be removed quickly.

We believe that a major improvement to the design would be a much smaller nozzle that directs more airflow locally.

Cost

The bill of materials for our final product:
- $88.95 for handheld vacuum, from which we used the impeller, motor, and battery
- ~$5 for the PLA used to print structural components
- ~$10 for aluminum mesh

Our design has an total cost of ~ $100. Considering that severe SLF infestations can reduce grape yield by up to 90%, this relatively low-cost solution could be a practical investment for protecting vineyard productivity.

This price is similar to the cost of spraying insecticide for a one acre field.

Design Feasibility + Improvements to Be Made

The device is not as efficient as desired, but it has potential nonetheless. However, given the size of the device and the amount of SLF that the device can remove from a plant at once, this design might work better when taken in one of the following directions.

The first is a scaled-up version with a more powerful motor and impeller system to collect more SLF at once. This design would also mean that the device could be used for longer periods before the SLF needs to be cleared out.

The second direction is to adapt the device for handheld use at the current scale. The current system can remove a few SLF at a time and could be operated efficiently by hand. The handheld version would work well on smaller vineyards.

A third possible direction is to connect the device to an actuator attached to the tractor. The actuator would enable the device to be used along the entire length of the vine, while reducing the nozzle size to increase suction capacity.